

Indian Institute of Technology Guwahati
Department of Physics
PH102
Tutorial-5

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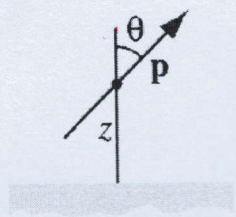
1. For some spherically symmetric distribution of charges, the potential on the surface of a sphere of radius R is given to be V_0 . The sphere does not contain any charge inside it. Calculate the potential in the regions $r > R$ and $r < R$.
2. A uniform line charge λ is placed on an infinite straight wire, a distance d above a grounded conducting plane. The wire runs parallel to the x -axis and directly above it, and the conducting plane is the xy plane.
 - (a) Find the potential in the region above the plane.
 - (b) Find the charge density σ induced on the conducting plane.
 - (c) Find the total induced charge on a strip of width w parallel to the y -axis.

3. According to quantum mechanics, the electron cloud for a hydrogen atom in the ground state has a charge density

$$\rho(r) = \frac{q}{\pi a^3} e^{-2r/a}$$

where q is the charge of the electron and a is the Bohr radius. Find the atomic polarizability of such an atom. [Hint: First calculate the electric field of the electron cloud, $E_e(r)$; then expand the exponential, assuming $r \ll a$.]

4. A dipole $\mathbf{p}$ is situated a distance z above an infinite grounded conducting plane as shown in the adjacent figure. The dipole makes an angle θ with the perpendicular to the plane. Find the torque on $\mathbf{p}$. If the dipole is free to rotate, in what orientation will it come to rest?



5. A sphere of radius R carries a polarization $\mathbf{P}(\mathbf{r}) = k\mathbf{r}$, where k is a constant and $\mathbf{r}$ is the vector from the centre.
 - (a) Calculate the bound charges σ_b and ρ_b .
 - (b) Find the field inside and outside the sphere.

Sol 1

The distribution is spherically symmetric and therefore there is only radius dependence. For $r < R$, there is no charge distribution inside the sphere $\therefore$ we can use the Laplace's eqn. in spherical polar coordinates.

$$\therefore \nabla^2 V = 0$$

$$\Rightarrow \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) = 0 \quad \text{--- (1)}$$

$$\Rightarrow r^2 \frac{\partial V}{\partial r} = C_1$$

$$\Rightarrow \frac{\partial V}{\partial r} = \frac{C_1}{r^2}$$

$$\Rightarrow V(r) = -\frac{C_1}{r} + C_2 \quad \text{--- (2)}$$

Now V must be finite within the sphere

$$\therefore C_1 = 0 \quad \text{--- (3)}$$

The boundary condition at $r=R$ is

$$V(R) = V_0 \quad \text{--- (4)}$$

$\therefore$ Using (3) & (4) in (2), we get

$$C_2 = V_0 \quad \text{--- (5)}$$

$$\Rightarrow V(r) = V_0 \quad \text{--- (6)}$$

Eq. (2) also holds for $r > R$ (no charge distribution) with the boundary conditions

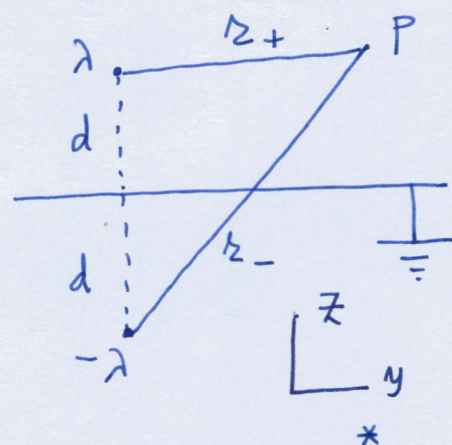
$$\left. \begin{array}{l} V = V_0 \text{ at } r = R \\ V = 0 \text{ at } r = \infty \end{array} \right\} \quad \text{--- (7)}$$

$\therefore$ using (7) in (2), we get

$$C_2 = 0 \quad \text{and} \quad C_1 = -V_0 R \quad \text{--- (8)}$$

$$\therefore V(r) = V_0 \frac{R}{r} \quad \text{--- (9)}$$

Sol. 2 (a) Using the method of images, consider a new configuration as shown in the figure. The potential at a point P is



$$V(y, z) = -\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_+}{s_0}\right) + \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_-}{s_0}\right)$$

$$= \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_-}{r_+}\right) \quad \text{--- (1)}$$

$$r_- = \sqrt{y^2 + (z+d)^2} \quad \text{and} \quad r_+ = \sqrt{y^2 + (z-d)^2} \quad \text{--- (2)}$$

$$\therefore V(y, z) = \frac{2\lambda}{4\pi\epsilon_0} \ln\left[\frac{y^2 + (z+d)^2}{y^2 + (z-d)^2}\right] \quad \text{--- (3)}$$

But it satisfies the boundary conditions
 $V(0,0) = 0$ and $V(y, z) \rightarrow 0$ as $y^2 + z^2 \gg d^2$.

$\therefore$ According to the uniqueness theorem, eq. (3) is the solution of the original problem in the region $z \geq 0$.

(b) We know that

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n} = -\epsilon_0 \frac{\partial V}{\partial z} \Big|_{z=0} \quad \text{--- (4)}$$

$$\frac{\partial V}{\partial z} = \frac{2\lambda}{4\pi\epsilon_0} \left\{ \frac{y^2 + (z-d)^2}{y^2 + (z+d)^2} \right\} \left\{ \frac{(y^2 + (z+d)^2) 2(z+d)}{[y^2 + (z+d)^2]^2} - \frac{(y^2 + (z-d)^2) 2(z-d)}{[y^2 + (z-d)^2]^2} \right\}$$

* For the individual line charge distribution,

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{S}. \quad \therefore V(s) = -\int_{s_0}^s \vec{E} \cdot d\vec{s} = -\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s}{s_0}\right).$$

$s_0 \rightarrow$ reference point. It must be common for any no. of charge distributions.

$$\therefore \left. \frac{\partial V}{\partial z} \right|_{z=0} = \frac{2\lambda}{4\pi\epsilon_0} \left[\frac{(y^2+d^2)(+2d) + (2d)(y^2+d^2)}{(y^2+d^2)^2} \right]$$

$$= + \frac{\lambda d}{\pi\epsilon_0} \left(\frac{1}{y^2+d^2} \right) \quad \text{--- (5)}$$

$$\therefore \sigma(y, z) = - \frac{\lambda d}{\pi} \left(\frac{1}{y^2+d^2} \right) \quad \text{--- (6)}$$

Sol.3 The field due to the electron cloud is

$$\vec{E}_e = \frac{q_{enc.}}{4\pi\epsilon_0 r^2} \hat{r} \quad \text{--- (1)}$$

$$\text{Where, } q_{enc} = \int_V \rho dz \quad \text{--- (2)}$$

$$\text{Now, } \rho(r) = \frac{q}{\pi a^3} e^{-2r/a}$$

$$\therefore q_{enc.} = \frac{q}{\pi a^3} \int_0^r e^{-2r/a} r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \quad \text{--- (3)}$$

$$= \frac{4q}{a^3} \int_0^r r^2 e^{-2r/a} dr \quad \text{--- (4)}$$

$$= \frac{4q}{a^3} \left[-\frac{a}{2} r^2 e^{-2r/a} \right]_0^r + \frac{a}{2} \int_0^r 2r e^{-2r/a} dr \right]$$

$$= \frac{4q}{a^3} \left[-\frac{a}{2} r^2 e^{-2r/a} + \left\{ 2\left(\frac{a}{2}\right) \left[\left(-\frac{a}{2}\right) r e^{-2r/a} \right]_0^r + \frac{a}{2} \int_0^r e^{-2r/a} dr \right\} \right]$$

$$= \frac{4q}{a^3} \left[-\frac{a}{2} r^2 e^{-2r/a} - \frac{a^2}{2} r e^{-2r/a} - a \left(\frac{a^2}{4} \right) e^{-2r/a} \right]_0^r$$

$$= \frac{4q}{a^3} \left[-\frac{a}{2} r^2 e^{-2r/a} - \frac{a^2}{2} r e^{-2r/a} - \frac{a^3}{4} e^{-2r/a} + \frac{a^3}{2} \right]$$

$$= + \frac{4q}{a^3} \left(\frac{a}{2} \right) \left[\frac{a^2}{2} - e^{-2r/a} \left(r^2 + ar + \frac{a^2}{2} \right) \right]$$

$$\text{--- (5)}$$

The external field separates the nucleus and the electron cloud by a distance d . At this distance external field balances the internal field ($r=d$).

$$\therefore \vec{E}_e = + \frac{2q}{a^2} \left(\frac{1}{4\pi\epsilon_0 d^2} \right) \left[\frac{a^2}{2} - e^{-2d/a} \left(d^2 + ad + \frac{a^2}{2} \right) \right] = \vec{E} \quad \text{--- (6)}$$

Now,

$$e^{-2d/a} = 1 - \left(+ \frac{2d}{a} \right) + \frac{1}{2} \left(\frac{2d}{a} \right)^2 + \frac{1}{3!} \left(- \frac{2d}{a} \right)^3 + \dots$$

$$= 1 - \frac{2d}{a} + \frac{1}{2} \left(\frac{2d}{a} \right)^2 - \frac{4}{3} \left(\frac{d}{a} \right)^3 + \dots \quad \text{--- (7)}$$

$$\therefore \vec{E}_e = \frac{q}{4\pi\epsilon_0 d^2} \left[1 - \left(\frac{2d^2}{a^2} + \frac{2d}{a} + 1 \right) \left(1 - \frac{2d}{a} + \frac{1}{2} \left(\frac{2d}{a} \right)^2 - \frac{4}{3} \left(\frac{d}{a} \right)^3 + \dots \right) \right] \hat{r}$$

$$= \frac{q}{4\pi\epsilon_0 d^2} \left[1 - \left\{ 1 + \frac{2d}{a} + \frac{2d^2}{a^2} \right\} + \frac{2d}{a} \left\{ 1 + \frac{2d}{a} + \frac{2d^2}{a^2} \right\} - \frac{1}{2} \left(\frac{2d}{a} \right)^2 \left(1 + \frac{2d}{a} + \frac{2d^2}{a^2} \right) + \frac{4}{3} \left(\frac{d}{a} \right)^3 \left\{ 1 + \frac{2d}{a} + \frac{2d^2}{a^2} \right\} + \dots \right] \hat{r}$$

$$= \frac{q}{4\pi\epsilon_0 d^2} \left[1 - \left(1 + \frac{2d}{a} + \frac{2d^2}{a^2} \right) + \frac{2d}{a} + \left(\frac{2d}{a} \right)^2 + \frac{2d}{a} \left(\frac{2d^2}{a^2} \right) - \frac{1}{2} \left(\frac{2d}{a} \right)^2 - \frac{1}{2} \left(\frac{2d}{a} \right)^3 - \frac{1}{2} \left(\frac{2d}{a} \right)^2 \left(\frac{2d^2}{a^2} \right) + \frac{4}{3} \left(\frac{d}{a} \right)^3 - \dots \right] \hat{r}$$

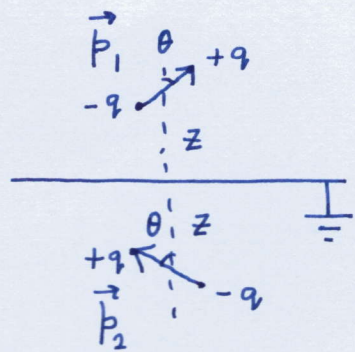
$$= \frac{q}{4\pi\epsilon_0 d^2} \left[1 - 1 - \frac{2d}{a} - \frac{2d^2}{a^2} + \frac{2d}{a} + \frac{4d^2}{a^2} + \frac{4d^3}{a^3} - \frac{2d^2}{a^2} - \frac{4d^3}{a^3} + \frac{4d^3}{3a^3} - \dots \right] \hat{r} \quad \text{--- (8)}$$

$$\sim \frac{4qd \hat{r}}{4\pi\epsilon_0(3a^3)} = \vec{E}$$

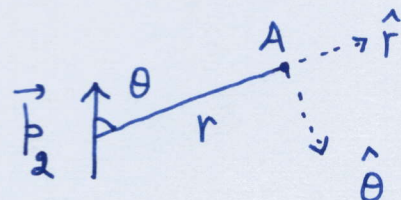
$$\text{or } \vec{p} = (3\pi\epsilon_0 a^3) \vec{E} \quad \text{--- (9)}$$

$$\therefore \alpha = 3\pi\epsilon_0 a^3 \quad \text{--- (10)}$$

Sol.4 We use the method of images & thus propose a new configuration as shown. To calculate the torque on $\vec{p}_1$, we first calculate the electric field due to the dipole $\vec{p}_2$ at the location of $\vec{p}_1$.



Consider the orientation as shown in the adjacent figure for $\vec{p}_2$. The dipole field at A is

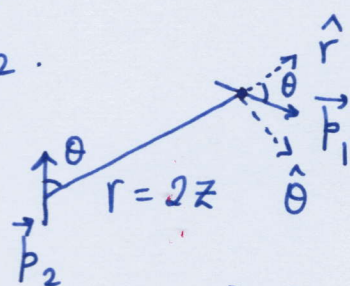


$$\vec{E}_2 = \frac{p_2}{4\pi\epsilon_0 r^3} \{ 2\cos\theta \hat{r} + \sin\theta \hat{\theta} \} \quad \text{--- (1)}$$

$\vec{p}_1$ experiences a torque due to $\vec{E}_2$.

$$\therefore \vec{N} = \vec{p}_1 \times \vec{E}_2 \quad \text{--- (2)}$$

$$\vec{p}_1 = p_1 \cos\theta \hat{r} + p_1 \sin\theta \hat{\theta} \quad \text{--- (3)}$$



$$(\vec{p}_1 = \vec{p}_2 = \vec{p})$$

and we get (using (1) & (3) in (2))

$$\vec{N} = (p_1 \cos\theta \hat{r} + p_1 \sin\theta \hat{\theta}) \times \frac{p_2 (2\cos\theta \hat{r} + \sin\theta \hat{\theta})}{4\pi\epsilon_0 (2z)^3}$$

$$= \frac{p_1 p_2}{32\pi\epsilon_0 z^3} (\cos\theta \sin\theta \hat{\phi} - 2\cos\theta \sin\theta \hat{\phi})$$

$$\text{or, } \vec{N} = - \frac{p^2 \sin\theta \cos\theta}{32\pi\epsilon_0 z^3} \hat{\phi} \quad \text{--- (4)}$$

If dipole is free to rotate, then $\vec{N} = 0$

$$\therefore \sin\theta \cos\theta = 0 \Rightarrow \theta = n\pi \text{ or } m\frac{\pi}{2} \quad \left. \begin{array}{l} \text{defines} \\ \text{the range} \\ \text{for } \theta. \end{array} \right\}$$

Thus, either $\theta = 0; \uparrow$ or $\theta = \pi; \downarrow$ gives the final orientation.

Sol.5 We have

$$\vec{P}(\vec{r}) = k\vec{r} = kr\hat{r} \quad \text{--- (1)}$$

$$\therefore \sigma_b = \vec{P} \cdot \hat{n} = kr\hat{r} \cdot \hat{r} = kR \quad (\text{at } r=R) \quad \text{--- (2)}$$

$$\begin{aligned} \rho_b &= -\vec{\nabla} \cdot \vec{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 P_r) = -\frac{1}{r^2} \frac{\partial}{\partial r} (kr^3) \\ &= -3k \quad \text{--- (3)} \end{aligned}$$

Equations (2) and (3) indicates that the sphere has a uniform charge distribution which is surrounded by a thin shell having a surface charge density σ_b .

We know that,

$$\vec{E} = \frac{\rho_b \vec{r}}{3\epsilon_0} ; r < R \quad \text{--- (4)}$$

$\therefore$ Using (3) in (4), we get

$$\vec{E} = -\frac{k\vec{r}}{\epsilon_0} ; r < R \quad \text{--- (5)}$$

$$\text{For } r > R, \quad \vec{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r} \quad \text{--- (6)}$$

$$\begin{aligned} \text{Now, } q &= \int \sigma_b da + \int \rho_b d\tau \quad \text{--- (7)} \\ &= \int_0^\pi \int_0^{2\pi} kR R^2 \sin\theta d\theta d\phi + (-3k) \int_0^R \int_0^\pi \int_0^{2\pi} r^2 \sin\theta dr d\theta d\phi \\ &= 4\pi kR^3 - 3k \left(\frac{4}{3} \pi R^3 \right) = 0 \quad \text{--- (8)} \end{aligned}$$

$$\therefore \vec{E} = 0 ; r > R \quad \text{--- (9)}$$